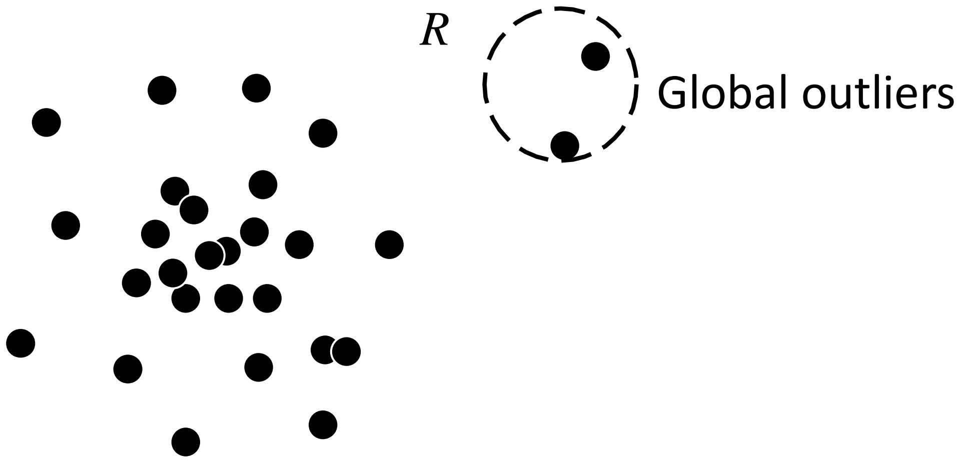
Three types of Outliers

- Global outliers
- Contextual (or conditional) outliers
- Collective outliers

Global Outliers

- A data object is a global outlier (or point anomaly) if it deviates significantly from the rest of the data set
  - Global outliers are the simplest type of outliers
- Most outlier detection methods are aimed at finding global outliers



Detecting Global Outliers

- It is critical to find an appropriate measurement of deviation with respect to the application in question
- Global outlier detection is important in many applications
  - Example: intrusion detection in computer networks
  - Example: trading transaction auditing systems

Contextual Outliers

- In a given data set, a data object is a contextual (or conditional) outlier if it deviates significantly with respect to a specific context of the object
- Example: "It is 86 degrees today" – Is it exceptional and thus an outlier?
  - Yes, if today is in winter in Toronto, but no if today is in summer in Toronto
- In contextual outlier detection, the context has to be specified as part of the problem definition

Contextual versus Behavioral Attributes

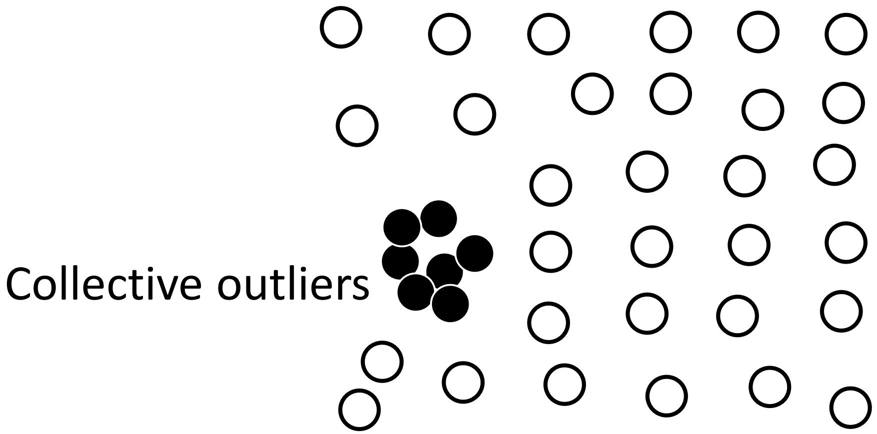
- In contextual outlier detection, the attributes of the data objects in question are divided into two groups
- The contextual attributes of a data object define the object's context
  - In the temperature example, the contextual attributes may be date and location
- The behavioral attributes define the object's characteristics, and are used to evaluate whether the object is an outlier in the context to which it belongs
  - In the temperature example, the behavioral attributes may be the temperature, humidity, and pressure
- Whether a data object is a contextual outlier depends on not only the behavioral attributes but also the contextual attributes

Contextual Outliers vs. Global and Local Outliers

- Contextual outliers are a generalization of local outliers
  - An object in a data set is a local outlier if its density significantly deviates from the local area in which it occurs
- Global outlier detection can be regarded as a special case of contextual outlier detection where the set of contextual attributes is empty – global outlier detection uses the whole data set as the context
- Contextual outlier analysis provides flexibility to users in that one can examine outliers in different contexts, which can be highly desirable in many applications
- The quality of contextual outlier detection in an application depends on the meaningfulness of the contextual attributes, in addition to the measurement of the deviation of an object to the majority in the space of behavioral attributes

Collective Outliers

- Given a data set, a subset of data objects forms a collective outlier if the objects as a whole deviate significantly from the entire data set, while importantly the individual data objects may not be outliers
- Example: a student sick case in a class may not be considered outlying, but it is outlying if 10% of students are sick on a single day



Applications of Collective Outliers

- In intrusion detection, a denial-of-service package from one computer to another is considered normal, and not an outlier at all
  - However, if several computers keep sending denial-of-service packages to each other, they as a whole should be considered as a collective outlier
  - The computers involved may be suspected of being compromised by an attack
- A stock transaction between two parties is considered normal
  - However, a large set of transactions of the same stock between two or a small number of parties in a short period are collective outliers because they may be evidence of some people manipulating the market
- In collective outlier detection we have to consider not only the behavior of individual objects, but also that of groups of objects
- To detect collective outliers, we need background knowledge of the relationship among data objects, such as distance or similarity measurements between objects

Comparison among Multiple Types of Outliers

- A data set can have multiple types of outliers
- An object may belong to more than one type of outlier
- Different outliers may be used in various applications or for different purposes
- Global outlier detection is the simplest
- Context outlier detection requires background information to determine contextual attributes and contexts
- Collective outlier detection requires background information to model the relationship among objects to find groups of outliers